Task 1: Running Basic Linux Commands

I ran the following Linux commands in the terminal and here is what I observed:

---

1. `ls`

The `ls` command is used to list all the files and folders present in the current directory. When I ran this command, it displayed all the folder names on the screen. This command is helpful when we want to know what files or folders are available in a location.

---

2. `pwd`

The `pwd` command stands for Print Working Directory. It shows the full path of the directory we are currently working in.

---

3. `cd`

The `cd` command stands for Change Directory. It is used to move from one folder to another folder. For example, if I want to go into the tmp folder, I will type `cd /tmp`. If I want to go back to the previous folder, I can type `cd ..`. This command helps us to navigate through the Linux file system.

---

4. `cat`

The `cat` command stands for 'concatenate'. It is used to display the contents of a file directly on the terminal screen. For example, if there is a file called `sample.txt`, typing `cat sample.txt` will print everything written inside that file.

Task 2: Directory Exploration

──(aswathy㉿kali)-[/var/log]

└─$ ls

alternatives.log boot.log.4 dpkg.log.2.gz lightdm notus-scanner speech-dispatcher alternatives.log.1 boot.log.5 dpkg.log.3.gz   macchanger.log   openvpn   stunnel4

apache2   boot.log.6 fontconfig.log macchanger.log.1.gz postgresql   sysstat

apt   boot.log.7 gvm   macchanger.log.2.gz private   wtmp

boot.log   btmp   inetsim   macchanger.log.3.gz README   Xorg.0.log

boot.log.1   btmp.1   installer   macchanger.log.4.gz redis   Xorg.0.log.old

boot.log.2   dpkg.log   journal   mosquitto   runit

boot.log.3   dpkg.log.1 lastlog   nginx   samba

Task 3: File Creation

──(aswathy㉿kali)-[~]

└─$ cd Downloads

┌──(aswathy㉿kali)-[~/Downloads]

└─$ touch test.txt

┌──(aswathy㉿kali)-[~/Downloads]

└─$ echo "Hello World"

Hello World

┌──(aswathy㉿kali)-[~/Downloads]

└─$ cat test.txt

┌──(aswathy㉿kali)-[~/Downloads]

└─$ echo "Hello World" > test.txt

┌──(aswathy㉿kali)-[~/Downloads]

└─$ cat test.txt

Hello World

LEVEL 2 – INTERMEDIATE

Task 4: Log Reading

1. I opened the **dpkg.log** file and I could see **package installation and update logs**. These logs record every time a software package was installed, removed or updated on the system using the apt or dpkg command.

2. 2026-03-11 10:27:04 startup archives unpack

   2026-03-11 10:27:06 upgrade locales:all 2.41-12 2.42-13

   2026-03-11 10:27:06 status half-configured locales:all 2.41-12

   2026-03-11 10:27:06 status unpacked locales:all 2.41-12

Level -3 Advanced

Task 7

1. I tried to analyze the logs on my Kali Linux system. I first went to the **/var/log/** directory and checked what log files were available by typing **ls /var/log/**. When I listed the log files I found the following files were available on my system: dpkg.log, boot.log, alternatives.log, macchanger.log, Xorg.0.log, fontconfig.log and some others.

   When I tried to open the following files they were not present on my system:
   syslog was not found
   auth.log was not found
   So I could not run the grep command because the auth.log file did not exist on my system.

   Kali Linux does not create syslog and auth.log by default because it uses **systemd journald** for logging instead of the traditional rsyslog system. These files are commonly found in Ubuntu and Debian systems but not in Kali Linux unless rsyslog is manually installed.

   Since auth.log and syslog were not available I analyzed the **dpkg.log** file which was available on my system. This file contains records of all software installations and updates.

In the dpkg.log file I did not find any suspicious activity. All the entries showed normal package installations and updates that I had done myself. There were no unknown or unauthorized software installations recorded.

2. Since auth.log was missing on my system I was unable to check for failed login attempts or unauthorized access attempts. The log files that were available on my system only contained software and boot related information and nothing suspicious was found in them.

Bonus Question

If a system shows 50 failed login attempts in just 1 minute it is a very clear sign that the system is under a **Brute Force Attack**. This means an attacker is using an automated program or script to try hundreds of different password combinations very quickly in order to gain unauthorized access to the system.

This is not normal human behavior because no genuine user would fail to login 50 times in just one minute. It could also be a **Dictionary Attack** where the attacker is trying a list of commonly used passwords one by one using tools like **Hydra**

This situation is very dangerous especially if the system has a weak password because the attacker can crack it easily. To prevent this we should use strong passwords, install **Fail2Ban** to automatically block suspicious IP addresses, and regularly monitor the log files for unusual activity.